

VISVESVARAYA NATIONAL INSTITUTE OF TECHNOLOGY

Department of Electrical & Electronics Engineering

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DIGITAL DNA MATCHING & MUTATION DETECTION SYSTEM

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SECTION 1

Project Overview

DNA (Deoxyribonucleic Acid) is the molecular building block for life, providing the genetic codes that determine the structure and function of living things. The accuracy of DNA base pairing is essential for replication, transcription, and repair. Inaccurate DNA base pairing or mutations may result in genetic diseases, improper functioning of cells, and cancer.

The purpose of this project is to develop a hardware model using conventional TTL logic ICs that validates whether DNA base pairs are matching or not. Two models have been tested using Proteus 8 software and the hardware for the same is constructed:

- Model 1 – Single-Stranded DNA Matching:** It tests for matching bases between two single-stranded DNA, where the matching is based on identical bases (i.e., A-A, T-T) and mutations (i.e., A-T, G-C, A-C). A logic has been developed to count and display the number of mutations (position-wise).
- Model 2 – Double-Stranded DNA Matching:** It tests for Watson-Crick complementary base pairs (i.e., A-T, T-A, G-C, C-G) through a magnitude comparator and SOP logic. For this model, the mutation count logic is also developed, only in simulation.

SECTION 2

Binary Encoding Scheme

MODEL 1:

Each single-stranded DNA base is assigned a unique coded Symbol. A base is therefore represented by a 2-bit binary code. The encoding for Single-Stranded DNA is:

Base	Symbol	2-bit Code
Adenine	A	00
Thymine	T	01
Guanine	G	10
Cytosine	C	11

For single-stranded DNA, the logic works as: When 2-bit of reference DNA base equals 2-bit of Sample DNA base (i.e., A-A, T-T, C-C, G-G), then it is a **Match** case, for the rest of the combinations (i.e., A-T, T-A, C-A), it is a **Mutation** case. And the counter will increase in each case of Mutation and will be displayed on the Seven Segment Anode Display.

MODEL 2:

Each Double-stranded DNA base is assigned a unique pair of coded symbols. A base is therefore represented by a 4-bit binary code. The encoding for Double-Stranded DNA is:

DNA Base	2-bit Code	Complement Base	Complement Code
Adenine (A)	00	Thymine (T)	01
Thymine (T)	01	Adenine (A)	00
Guanine (G)	10	Cytosine (C)	11
Cytosine (C)	11	Guanine (G)	10

When a 4-bit reference DNA matches with the 4-bit Sample DNA, the comparator gives an output 1 for Matching. At the Same time, it checks whether each DNA base is present with it's complementary base. If both conditions are true, then the final output becomes 1 and it is a **Match** case. If any one of the conditions is not true, it is detected as a **Mutation** case.

SECTION 3

Model 1 — Single-Stranded DNA Matching

This model compares two individual DNA bases — a **Reference strand** (R1 R0) and a **Sample strand** (S1 S0). A match is declared when both bases are identical (e.g., A–A, T–T, G–G, C–C). When a mismatch occurs, it is counted as a **mutation**, and the 7-segment display increments the mutation count.

3.1 Logic Design

The 2-bit comparison is implemented using **74HC86 XOR gates**. For each bit position, an XOR gate produces logic HIGH when the bits differ. If both XOR outputs are LOW (i.e., R1=S1 AND R0=S0), the bases match. An OR gate combines the XOR outputs: if the OR output is HIGH, a mismatch (mutation) has occurred.

- **LED indicator:** **Red** – Matching; **Green**- Mutation
- **Mutation signal triggers:** 74LS74 edge detector → 74LS90 counter → 74LS47 → 7-segment display
- NE555 timer (astable mode) provides the sampling clock to the D flip-flop
- Resistors & capacitor of appropriate value is used to provide a clock pulse for desired duration of time.

3.2 ICs Used in Model 1

IC / Component	Role	Quantity
Toggle Switch	To provide binary input	4
74HC86	XOR gates for bit-wise comparison	2
74HC32	OR gates to combine XOR outputs	1
74HC04	Inverter for logic conditioning	1
74LS74	D flip-flop for edge detection	1
74LS90	BCD decade counter for mutation count	1
74LS47	BCD to 7-segment decoder	1
74LS08	AND gates for combining conditions	1
NE555	Astable oscillator — sampling clock	1
7-seg display	Displays mutation count	1
LED-RED	DNA match indicator	1
LED-GREEN	Mutation indicator	1
Resistor(10k,100k); Capacitor (100uF)	Frequency / timing adjustment	2
Resistors (330ohm)	Current limiting resistors for LEDs	1
Resistors (330ohm)	Current limiting resistors for LEDs	2
Battery (5V)	Power Supply	1

3.3 Proteus Schematic — Model 1

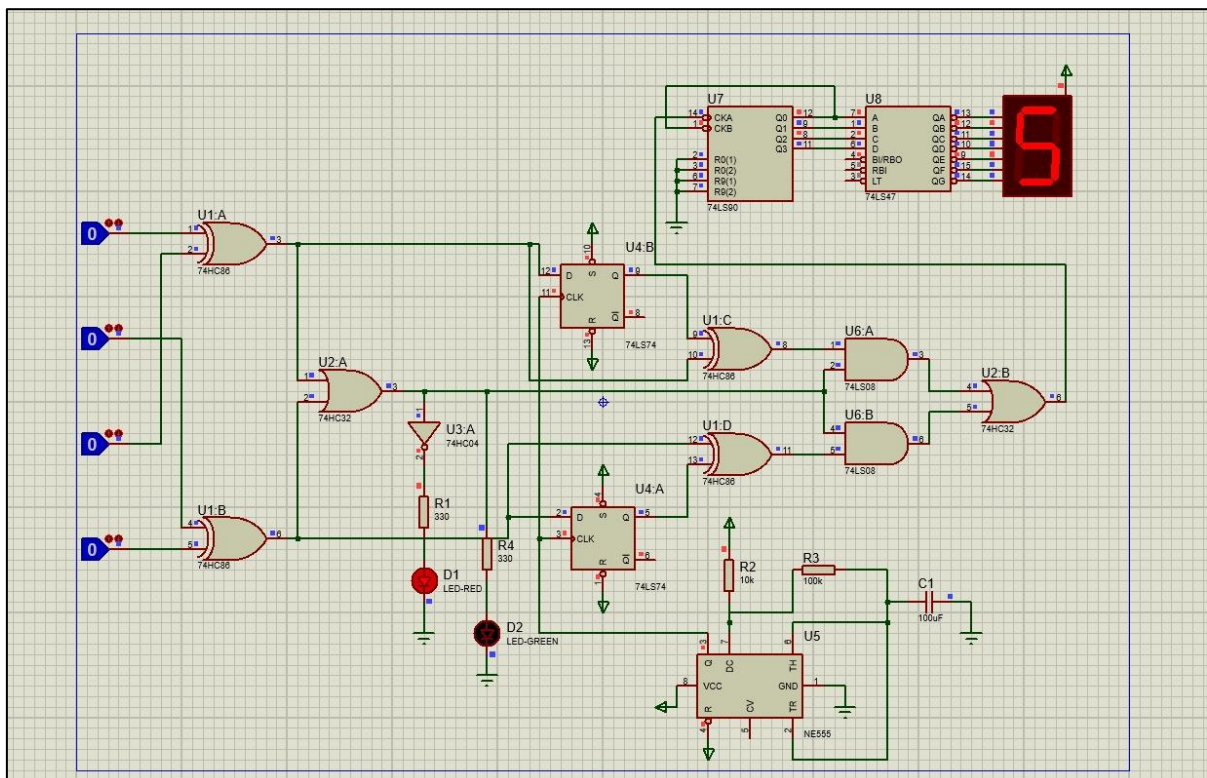


Figure 1 — Proteus Schematic: Single-Stranded DNA Matching System

SECTION 4

Model 2 — Double-Stranded DNA Matching

This model does the full double-stranded DNA base-pair verification. Two independent 4-bit inputs represent a complete base pair each — the **Left strand** (R3–R2) and the **Right strand** (R1–R0) for reference DNA; the **Left strand** (S3–S2) and the **Right strand** (S1–S0) for sample DNA. A valid Watson-Crick match is detected only when:

- 1. Both 4-bit inputs are **identical** (verified by the 74LS85 comparator), AND
- 2. The 4-bit value is one of exactly **4 valid patterns**: 0001 (A–T), 0100 (T–A), 1011 (G–C), 1110 (C–G).

On a confirmed **DNA MATCH**: a red LED illuminates and the 7-segment counter increments (in case of simulation only), tracking the total number of matches observed.

4.1 Two-Condition Match Logic

Condition 1 — Equality: 74LS85 magnitude comparator (A=B output, pin 6) checks Left == Right.

Condition 2 — Valid Pattern: A combinational network of logic gates checks if the value is 0001, 0100, 1011, or 1110.

Final AND: DNA MATCH = EQUAL AND VALID PATTERN

4.2 ICs Used in Simulation 2

IC/Component	Role	Quantity
74LS85	4-bit magnitude comparator	1
74LS04	Hex inverter	1
74LS08	Quad AND — SOP first/second level & final AND gate	4
74LS32	Quad OR — combines 4 pattern outputs into VALID PATTERN	1
74LS74	D flip-flop — rising-edge detection on DNA_MATCH	1
74LS90	Decade counter — sampling clock for edge detector	1
74LS47	BCD to 7-segment decoder	1
7-seg display	Displays match count (0–9)	1
LED-RED	Match indicator LED	1
Resistor (330 ohm)	Current limiting resistor for LED	1
Battery (5V)	Power Supply	1
Toggle Switch	Binary input switches	8

4.3 Proteus Schematic — Simulation 2

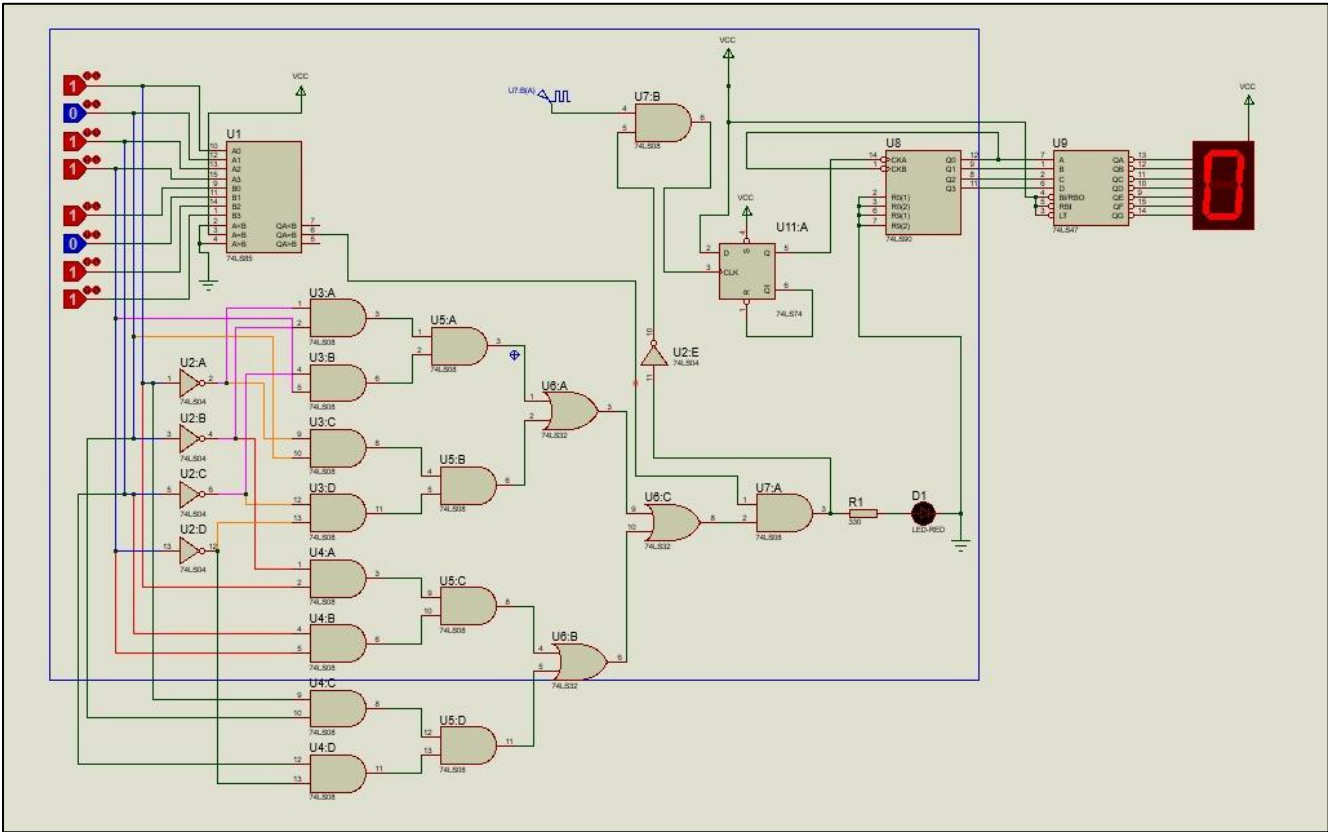


Figure 2 — Proteus Schematic: Double-Stranded DNA Matching System

SECTION 5

Truth Tables

5.1 Single-Stranded DNA — Truth Table (Model 1)

R1 R0 = Reference base; S1 S0 = Sample base. A match occurs when both 2-bit codes are identical.

R1	R0	Ref Base	S1	S0	Sample Base	Match?	Result
0	0	A	0	0	A	✓ MATCH	MATCH
0	1	T	0	1	T	✓ MATCH	MATCH
1	0	G	1	0	G	✓ MATCH	MATCH
1	1	C	1	1	C	✓ MATCH	MATCH
0	0	A	0	1	T	✗	MUTATION
0	0	A	1	0	G	✗	MUTATION
0	1	T	1	1	C	✗	MUTATION
1	0	G	1	1	C	✗	MUTATION

5.2 Double-Stranded DNA — Full Truth Table (Model 2, all 16 left-input values)

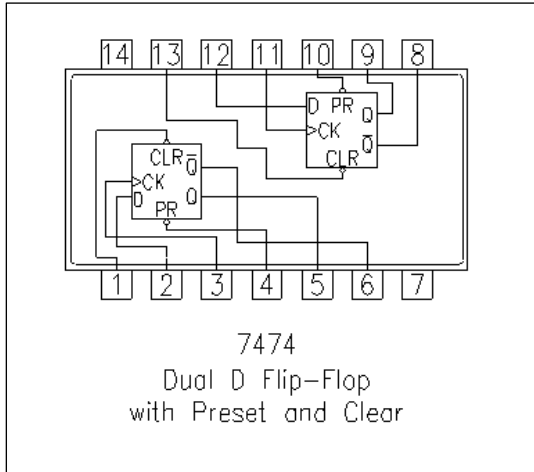
R3 R2 R1 R0 = Reference base; S3 S2 S1 S0 = Sample base. A match occurs when both 4-bit codes are identical which is checked by comparator. And it is a match only when bases are paired with their correct complementary bases.

R3	R2	R1	R0	Dec	Base Pair	Valid?	Result
0	0	0	0	0	A–A	No	MUTATION
0	0	0	1	1	A–T	Yes	MATCH
0	0	1	0	2	A–C	No	MUTATION
0	0	1	1	3	A–G	No	MUTATION
0	1	0	0	4	T–A	Yes	MATCH
0	1	0	1	5	T–T	No	MUTATION
0	1	1	0	6	T–C	No	MUTATION
0	1	1	1	7	T–G	No	MUTATION
1	0	0	0	8	G–A	No	MUTATION
1	0	0	1	9	G–T	No	MUTATION
1	0	1	0	10	G–G	No	MUTATION
1	0	1	1	11	G–C	Yes	MATCH
1	1	0	0	12	C–A	No	MUTATION
1	1	0	1	13	C–T	No	MUTATION
1	1	1	0	14	C–G	Yes	MATCH
1	1	1	1	15	C–C	No	MUTATION

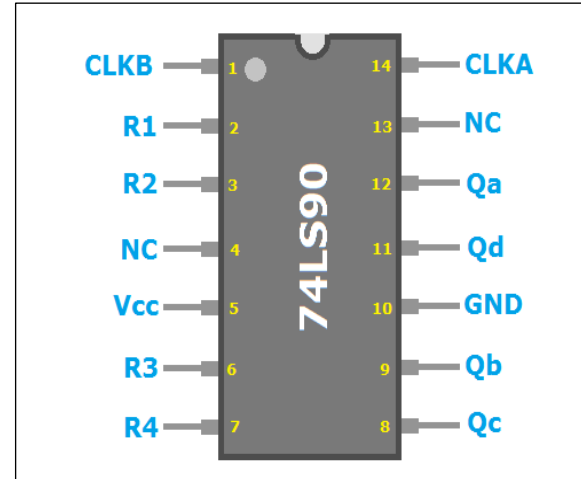
SECTION 6

INTERNAL STRUCTURES OF ICs USED

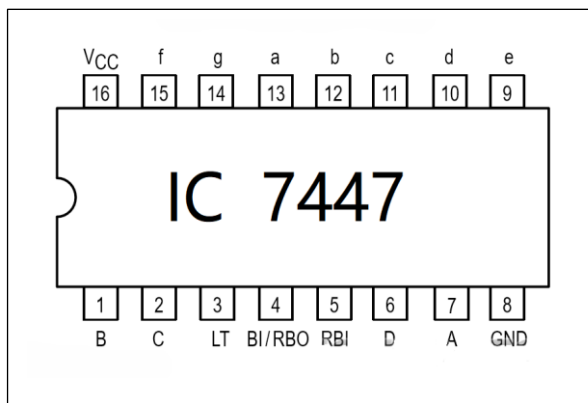
6.1 D Flip-Flop (74LS74)



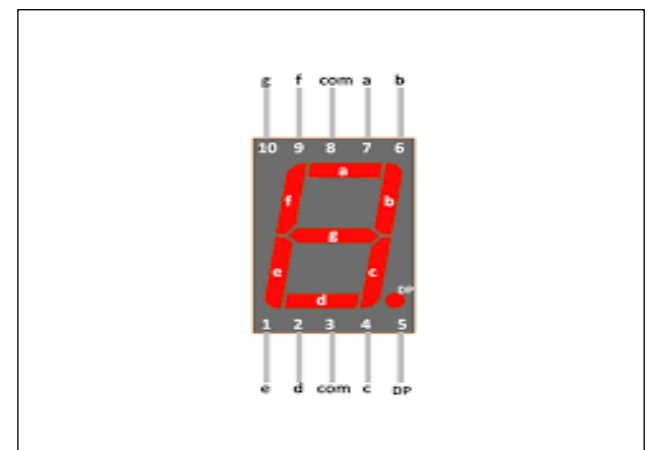
6.2 Decade Counter (74LS90)



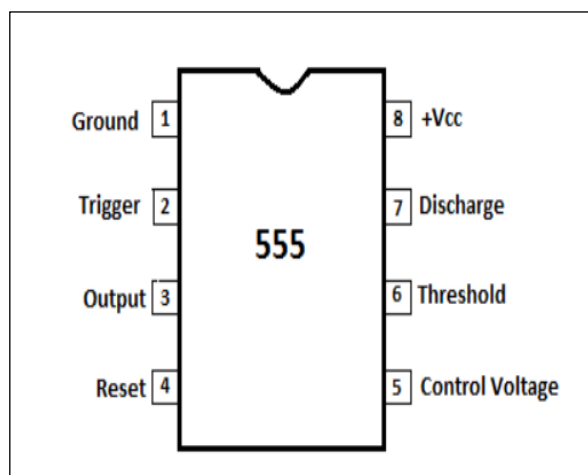
6.3 BCD to 7-segment decoder (74LS47)



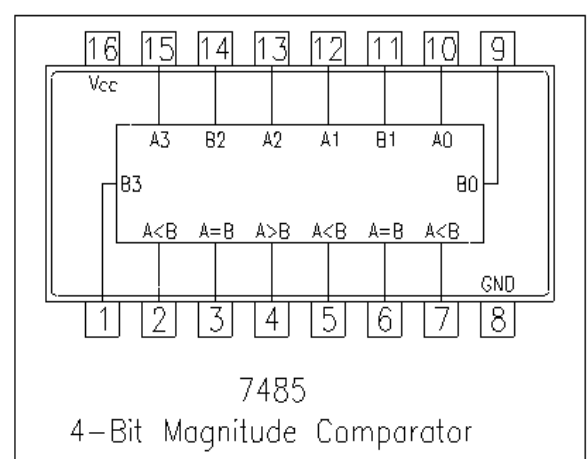
6.4 7-segment Anode Display



6.5 555-Timer (Astable mode)



6.6 Comparator (74LS85)



SECTION 7

WORKING

The system implements single-strand DNA matching and mutation detection using digital logic circuits.

1. DNA Base Comparison

The reference DNA bits (**R1, R0**) and sample DNA bits (**S1, S0**) are compared using XOR gates.

The pair (**R1, S1**) is applied to one XOR gate.

The pair (**R0, S0**) is applied to another XOR gate.

An XOR gate produces a HIGH output only when the inputs are different. Therefore:

Output = 0 → Bits match

Output = 1 → Bits do not match

The outputs of both XOR gates are then applied to an OR gate.

If any mismatch occurs, the OR gate output becomes HIGH.

This HIGH signal turns ON a **green LED**, indicating mismatch.

When both bits match, the OR gate output remains LOW. This output is inverted using a NOT gate:

The NOT gate output becomes HIGH when matching occurs

This drives a **red LED**, indicating mutation detection condition as per system logic

2. Mutation Detection Using Previous State

To detect mutations over time (i.e., compare current and previous states), D flip-flops are used.

Outputs from the initial XOR gates are given as inputs to two D flip-flops.

These flip-flops store the **previous state** of the comparison.

The stored outputs (Q) are then compared with the current XOR outputs using additional XOR gates:

If current and previous outputs differ, it indicates a **change in mutation state**

Thus, this stage detects **transitions in mutation**, not just static mismatches.

3. Controlled Trigger Generation

The outputs from the second stage XOR gates are passed through AND gates.

- Each AND gate have two inputs:
 1. Output of transition-detecting XOR
 2. Mutation detection signal (output of NOT gate from Stage 1)

This ensures:

- The signal is allowed to pass **only when a mutation is detected**
- Prevents false triggering when no mutation exists

4. Pulse Generation and Counting

- Outputs from both AND gates are combined using an OR gate.
- This OR gate output acts as a **clock pulse input** to the **74LS90 decade counter**.

Each valid mutation event generates one clock pulse, incrementing the counter.

The counter output (BCD format) is then fed to a **BCD to 7-segment decoder**, which drives a **7-segment display**.

5. Display of Mutation Count

The 7-segment display shows the **total number of detected mutations**. Each detected mutation increments the count by one, allowing real-time monitoring of mutation events.

A similar approach is extended for double-stranded DNA matching, where all valid base-pair combinations are considered and a comparator-based logic is used to verify correct pairing and detect mutations.

SECTION 8

Observed Results

- The single-stranded model works correctly for all matching cases (A-A, T-T, C-C, G-G) and also detects mutation for all mismatched cases, along with proper display of mutation counts.
- In double stranded model, all four valid Watson-Crick match patterns (0001, 0100, 1011, 1110) correctly triggered the red LED and incremented the 7-segment counter.
- All invalid same-base pairs (A-A, T-T, G-G, C-C) correctly held the LED OFF and left the counter unchanged.
- All mismatched inputs (Left $\neq$ Right) produced no match, even when one strand individually encoded a valid pattern.
- The 74LS85 equality check reliably prevented false matches for all mismatched input combinations.
- The D flip-flop (74LS74) edge detection ensured the counter incremented exactly once per match event, not continuously.
- The 7-segment display correctly showed BCD counts 0 through 9 for successive match events.
- The LED responses verify the working of DNA matching and mutation detection system.
- The system correctly counted mutation events and displayed the mutation count on the 7-segment.
- Hardware Implementations are done and verified as per the simulated circuits display via the NE555 clock and 74LS90 counter.
- Both simulations were fully verified and confirmed WORKING in Proteus 8 simulation.

SECTION 9

Challenges & Solutions

Defining the Match Condition Correctly

Initial design attempted direct $A=T$ bit-level comparison, which is incorrect. The 4-bit input encodes a complete pair, so the correct approach is an equality check (74LS85) combined with a valid-pattern SOP check, not a complementarity check.

4-input AND Gates with 2-input ICs

The 74LS08 provides only 2-input AND gates. Each 4-input AND was implemented in two stages: $(A \text{ AND } B)$ and $(C \text{ AND } D)$ computed separately, then ANDed together. This required additional IC packages but achieves correct logic.

Counter Incrementing Continuously on Held Match

A static HIGH on DNA MATCH caused the counter to count on every clock edge. Solved by using the 74LS74 D flip-flop as a one-shot edge detector — only the rising edge of DNA MATCH triggers a single count pulse.

74LS85 Cascade Input Configuration

Incorrect cascade wiring produced wrong comparison results. For standalone use: $A=B$ cascade input (pin 2) must be tied to VCC; $A>B$ (pin 4) and $A<B$ (pin 3) must be tied to GND.

NE555 Timer Frequency Calibration (Sim 1)

Initial 555 astable frequency was too high, causing multiple counts per mutation event. The values of resistors and capacitors were adjusted to bring the sampling rate within the flip-flop's setup/hold time requirements.

LED Polarity and Current Limiting

Early versions had the LED wired without a current limiting resistor or with reversed polarity. Resolved by adding a $330\ \Omega$ series resistor and verifying anode/cathode orientation in Proteus.

SECTION 10

Conclusion & Future Scope

This project successfully demonstrates a simulation and hardware implementation of DNA base-pair matching and mutation detection using standard ICs. Two distinct models were designed, verified, and documented:

- **Model 1** (Single-Stranded) identifies same-base matches using XOR-based comparison and counts mutations on a 7-segment display driven by a NE555 clock.
- **Model 2** (Double-Stranded) verifies all four Watson-Crick complement pairs using a 74LS85 comparator combined with SOP pattern logic, illuminating a red LED for match and incrementing a mutation counter on mismatched pairs.

The Proteus simulation confirmed full functional correctness across all tested input combinations. The edge-triggered counter ensures accurate match counting, and the 7-segment display provides a clear visual readout. The hardware implementation is done by referring the simulated circuit and the results have been verified.

Future Scope

- Extend to simultaneous multi-base-pair verification by cascading the pattern-detection blocks.
- This system can be further extended for real-time genomic analysis and integration with bioinformatics tools for advanced mutation detection.
- It includes scaling the design for longer DNA sequences and implementing it using FPGA or VLSI for high-speed applications.
- Incorporate full codon (3-base) detection for protein synthesis simulation.
- Add a serial interface to input DNA sequences programmatically.

SECTION 11

References

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- [7] Texas Instruments, *74LS47 BCD to 7-Segment Decoder Datasheet*, Texas Instruments Inc.
- [8] Lab center Electronics, *Proteus 8 Professional — ISIS Schematic Capture User Manual*.
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- [10] Alberts, B. et al., *Molecular Biology of the Cell*, 6th ed., Garland Science, 2014.